

UV-VISIBLE SPECTROSCOPY TO

DETERMINE THE NATURE OF BAND

GAP & ESTIMATION OF BAND-GAP

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Abstract.

In this experiment, UV-Visible spectroscopy was used to determine the band gaps of semiconductor thin films using Tauc method. The absorption spectra were analysed by plotting $(\alpha h\nu)^2$ v/s energy for direct band gap & $\sqrt{\alpha h\nu}$ v/s energy for indirect band gap materials. The band gap values obtained were (2.143 ± 0.003) eV for CdS, (2.207 ± 0.013) eV for ZnO, (1.925 ± 0.003) eV for ZnTe, (1.532 ± 0.001) eV for TiO_2 & an indirect band gap of (1.6350 ± 0.0007) eV for the given polymer. data. The results demonstrate the applicability of UV-Vis spectroscopy in determining the nature & magnitude of band gaps in electronic band structure of semiconductor materials.

Objective :-

- i) Estimation of the band gap for direct & indirect band gap materials using UV-Vis spectra.

Theory :-

Band gap in semiconductors is the minimum amount of energy required to excite an electron from the valence band to the conduction band. It is significant in determining the electrical & optical properties of materials.

① Direct Band Gap

In direct band gap semiconductors, the conduction band minimum & the valence band maximum occur at the same momentum (k). Therefore, e^- transitions occur directly by absorbing a photon without requiring a change in momentum. These materials exhibit strong absorption & are widely used in optoelectronics devices. (See Fig. 1 for reference)

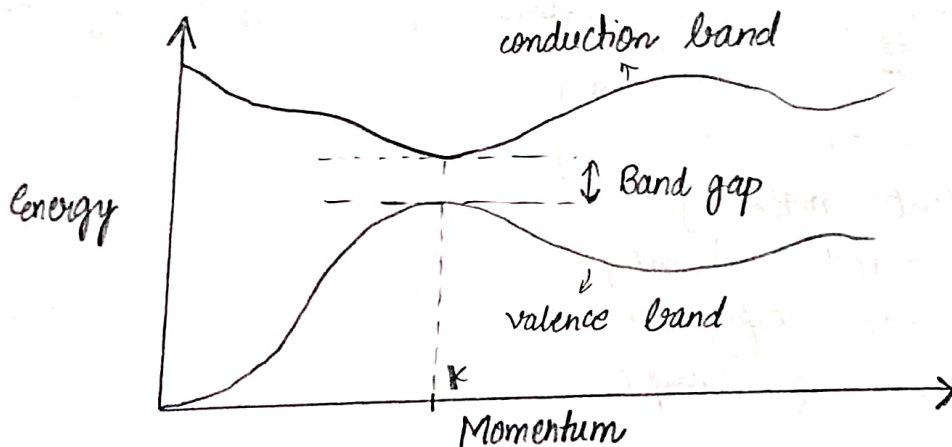


Fig 1: Band structure for direct band gap material.

② Indirect band gap

In indirect band gap semiconductors, the conduction band minimum & valence band maximum occur at different momenta (k). Hence, the e^- transitions require both a photon (for energy) and a phonon (for momentum conservation). This makes the

process less probable, resulting in weaker optical absorption.
(see Fig. 11)

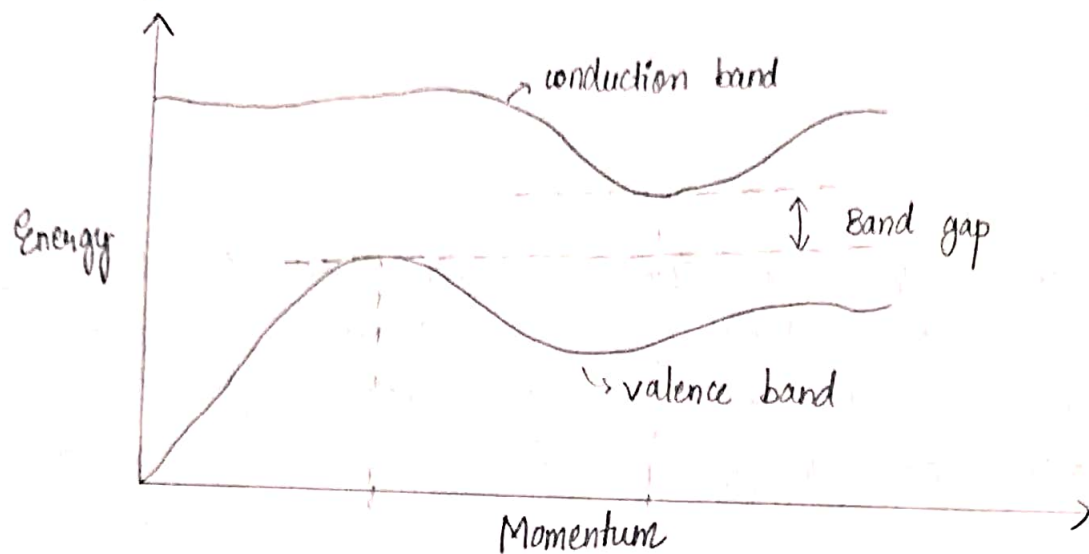


Fig 11:- Band structure for indirect band gap material.

* UV - Visible Spectroscopy :-

UV - Visible spectroscopy is used to measure the absorbance of a material as a function of wavelength.

→ Beer - Lambert's Law

The absorbance of light in a material is described by the relation,

$$I_T = I_0 e^{-\alpha d}$$

where I_0 = incident intensity,
 I_T = transmitted intensity
 α = absorption coefficient
 d = thickness of sample.

The absorbance A is given by,

$$A = \log_{10} \left(\frac{I_0}{I_T} \right)$$

From this, absorption coefficient can be calculated as,

$$\alpha = \frac{2.303 A}{d}$$

* Tauc Relation :-

The relationship b/w absorption coefficient & photon energy is given by Tauc equation :

$$\alpha = C(h\nu - E_g)^n$$

$h\nu$ = photon energy , C = constant

n = depends on ~~each~~ type of electronic transition:

- $n = 1/2$ for direct allowed transitions
- $n = 2$ for indirect allowed transitions.

For direct band gap materials, a plot of $(\alpha h\nu)^2$ v/s $h\nu$ is plotted. The linear region near the absorption edge is extrapolated to the energy axis to obtain the band gap.

For indirect band gap materials, a plot of $\sqrt{\alpha h\nu}$ v/s $h\nu$ is used. The intercept of the linear region on energy axis gives the band gap.

Direct band gap materials show a sharp rise in absorption near the band gap edge, whereas indirect band gap materials exhibit a gradual increase in the absorption spectrum.

Apparatus Required :-

- i) UV-Visible Spectrometer with light source
- ii) Computer with spectral analysis software
- iii) Sample holder
- iv) Thin-film semiconductor samples deposited on glass substrates
- v) Blank glass slides
- vi) Optical cables & detector unit
- vii) Power supply for spectrometer.

Math
02/04/26

Calculations :-

We used OriginLab for data analysis & plotting. We used the following relations

i) conversion of wavelength to energy : $E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$

ii) Absorbance (A) : $A = \log_{10}\left(\frac{I_0}{I_T}\right)$

iii) Absorption coefficient (α) : $\alpha = \frac{2.303A}{d}$ where d = thickness of sample.

We plot $(\alpha h\nu)$ v/s $h\nu$, $(\alpha h\nu)^2$ v/s $h\nu$ & $\sqrt{(\alpha h\nu)}$ v/s $h\nu$ for each of the sample & analyze the spectrum.

We identify the linear region in the plots ~~if they exhibit~~ of $(\alpha h\nu)^2$ v/s $h\nu$ & $\sqrt{\alpha h\nu}$ v/s $h\nu$ if they exhibit any. The provided samples CdS, ZnTe, TiO_2 & ZnO show a clear linear region in their $(\alpha h\nu)^2$ v/s $h\nu$

plots, so we classify them as direct band gap materials. They do not show a clear linear region in $\sqrt{\alpha h\nu}$ v/s $h\nu$ plots. A file for a polymer with data for indirect band gap material was provided & we perform an analysis on that by fitting the linear region of the $\sqrt{\alpha h\nu}$ v/s $h\nu$ plot.

From the attached plots, using the Tauc relation,

$$\alpha h\nu = A (h\nu - E_g)^n$$

$n = \frac{1}{2}$, for indirect transitions

$n = 2$, for indirect transitions.

For direct band gap.

$$\alpha h\nu = A (h\nu - E_g)^{\frac{1}{2}} \Rightarrow (\alpha h\nu)^2 = A^2 (h\nu - E_g)$$

We perform a linear fit in the linear region of $(\alpha h\nu)^2$ vs $h\nu$ plot

$$Y = mX + C.$$

$$X = h\nu = \text{Energy in eV}$$

$$Y = (\alpha h\nu)^2$$

put $Y = 0$ (X-intercept).

$$(\alpha h\nu)^2 = A^2 (h\nu - E_g)$$

$$\Rightarrow 0 = A^2 (h\nu - E_g)$$

$$\boxed{h\nu|_{\text{X-intercept}} = E_g}$$

Hence, the X-intercept energy gives the band gap. E_g .

For indirect band gap,

$$\alpha h\nu = A (h\nu - E_g \mp E_p)^2$$

$$\text{if } E_g \gg E_p, (\alpha h\nu)^{\frac{1}{2}} = A^{\frac{1}{2}} (h\nu - E_g)$$

We perform a linear fit of $(\alpha h\nu)^{\frac{1}{2}}$ vs $h\nu$ plot

$$\underline{\text{X-intercept}} :- \boxed{h\nu|_{\text{X-intercept}} = E_g}$$

From the attached plots, we obtain the band gap from the X-intercept of $(\alpha h\nu)^2$ vs $h\nu$ plot for direct band gap materials:

i) CdS sample,

$$E_g = (2.143 \pm 0.003) \text{ eV}$$

ii) ZnO sample,

$$E_g = (2.207 \pm 0.013) \text{ eV}$$

iii) for ZnTe sample, $E_g = (1.925 \pm 0.003) \text{ eV}$,

For given polymer data, we obtain indirect band gap from x-intercept of the linear region of $\sqrt{\alpha h\nu}$ v/s $h\nu$ plot,

iv) Polymer, $E_g = (1.6350 \pm 0.0007) \text{ eV}$

For TiO_2 , we observe an almost linear region in both $(\alpha h\nu)^2$ v/s $h\nu$ & $\sqrt{\alpha h\nu}$ v/s $h\nu$ plots. However, ~~since~~ from the x-intercepts of the linear regions, we identify the $(\alpha h\nu)^2$ v/s $h\nu$ plot linear region to be more meaningful & classify it as direct band gap material.

From the plot we obtain, for TiO_2 sample

$$E_g = (1.5327 \pm 0.001) \text{ eV}$$

Error Analysis :

① Linear fit using Least Squares Method :-

The linear least squares fitting fits the data points into an equation of the form $y = ax + b$.

The loss function minimised is,

$$L(a, b) = \sum_{i=1}^N (\hat{y}_i - (ax_i + b))^2$$

$\hat{y}_i$ = true value
 a, b = parameters

Choosing the optimal parameters a & b to minimize the loss function,

$$\frac{\partial L}{\partial a} = -2 \sum_{i=1}^N x_i (y_i - (ax_i + b)) = 0$$

$$\frac{\partial J}{\partial b} = -2 \sum_{i=1}^N (y_i - (ax_i + b))^2 = 0$$

$N = \text{no. of observations}$

$\{x_i, y_i\}_{i=1}^N = \text{Data points}$

$$\Rightarrow \begin{cases} a \sum_{i=1}^N x_i^2 + b \sum_{i=1}^N x_i = \sum_{i=1}^N x_i y_i \\ a \sum_{i=1}^N x_i + bN = \sum_{i=1}^N y_i \end{cases}$$

Solving the linear system with 2 equations & 2 unknowns, gives a & b .

Error in y , $\sigma_y = \sqrt{\frac{1}{(N-2)} \sum_{i=1}^N (y_i - (ax_i + b))^2}$

Error in slope (a), $\sigma_a = \sigma_y \sqrt{\frac{N}{\Delta}}$ $\Delta = N \sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2$

Error in intercept (b) $\sigma_b = \sigma_y \sqrt{\frac{\sum_{i=1}^N x_i^2}{\Delta}}$

(II) Error Propagation :-

we find X-intercept using,

$$X\text{-intercept} = -\frac{b}{a}$$

$$\left\{ \begin{array}{l} y = ax + b \\ @ \quad y = 0 \\ \quad x = -\frac{b}{a} \end{array} \right\}$$

$\therefore$ Error in X-intercept,

$$\sigma_{X\text{-int}} = \sqrt{\left(\frac{\partial(X\text{-int})}{\partial b} \cdot \sigma_b \right)^2 + \left(\frac{\partial(X\text{-int})}{\partial a} \cdot \sigma_a \right)^2}$$

$$= \sqrt{\left(-\frac{1}{a} \cdot \sigma_b \right)^2 + \left(\frac{b}{a^2} \cdot \sigma_a \right)^2} = \sqrt{\left(\frac{-\frac{b}{a} \cdot \sigma_b}{b} \right)^2 + \left(\frac{-\frac{b}{a} \cdot \sigma_a}{a} \right)^2}$$

$$\boxed{\sigma_{X\text{-int}} = (X\text{-int}) \sqrt{\left(\frac{\sigma_b}{b} \right)^2 + \left(\frac{\sigma_a}{a} \right)^2}}$$

We used ORIGIN which directly implements the fitting & provides us the X-intercept & its uncertainty.

III Percentage Deviation

From the literature values of band gap energy for given samples, we calculate the % deviation in each measurement.

$$\% \text{ deviation} = \frac{|E_{g \text{ exp.}} - E_{g \text{ lit.}}|}{E_{g \text{ lit.}}} \times 100 \%$$

Samples	$E_{g \text{ experimental}} \text{ (eV)}$	$E_{g \text{ literature}} \text{ (eV)}$	% deviation
i) CdS	(2.143 ± 0.003)	2.42	11.4%
ii) ZnO	(2.207 ± 0.013)	3.35	34%
iii) ZnTe	(1.925 ± 0.003)	2.21	12.8%
iv) TiO_2	(1.532 ± 0.001)	3.15	51.4%

Results & Discussion :-

The optical band gaps were calculated for thin film samples given using the Tauc relation. For direct band gaps X-intercept of linear region of $(\alpha h\nu)^2$ v/s $h\nu$ plot were used. However, for indirect band gaps, $\sqrt{\alpha h\nu}$ v/s $h\nu$ plots were used. The experimental band gap values along with their percentage deviation are :

Sample	Nature	$E_{g \text{ (exp.)}} \text{ (eV)}$	$E_{g \text{ (lit.)}} \text{ (eV)}$	% deviation
i) CdS	Direct	2.143 ± 0.003	2.42	11.4%
ii) ZnO	Direct	2.207 ± 0.013	3.35	34%
iii) ZnTe	Direct	1.925 ± 0.003	2.21	12.8%
iv) TiO_2	Direct	1.532 ± 0.001	3.15	51.4%
v) Polymer	Indirect	1.6350 ± 0.0007	-	-

The Tauc plots for all samples showed a region of linearity, indicating optical transitions corresponding to their respective bandgap nature. Direct band gap materials exhibited a sharp absorption edge & a well defined linear region in $(\alpha h\nu)^2$ plots. However, indirect band gap materials showed a more gradual absorption edge & their band gaps were estimated using $\sqrt{\alpha h\nu}$ plots, showing indirect transitions.

We observed a lot of deviation from literature values due to experimental uncertainties, instrumental noise, film thickness variation. Impurities & defects in samples may introduce errors in band gap calculations.

Overall, the experiment demonstrates the use of UV-Vis spectroscopy & Tauc analysis in determining the nature & magnitude of band gap in semiconducting materials.

Conclusion :-

In the experiment, UV-Vis spectroscopy was used to estimate the band gaps (optical) of semiconductor thin films using Tauc method. The Tauc plots enabled in the identification of nature of band gap & their estimation. Deviations were observed from literature values which can be attributed to experimental uncertainties & limitations (also sample impurity). Overall, the experiment demonstrates the effectiveness of UV-Vis spectroscopy in probing the electronic structure of semiconducting ~~non~~ materials.

Sources of Error

- i) Non-uniform thickness of thin films
- ii) Instrumental noise & fluctuations.
- iii) Surface contamination of samples & due to improper handling.
- iv) Sample impurities & defects. can affect band gap calculations.

